

Chapter 2

Attributes of a Project

Projects can be looked at in terms of their attributes

- Time frame
- Purpose
- Ownership
- Resources
- Roles
- Risks
- Assumptions
- Independent tasks
- Organisational changes
- Operating in an environment larger than project itself

Time Frame

Since a project is a temporary endeavour, certainly it must have a beginning and an end. Several projects start on a specific date while completion date is estimated. But on the other hand, some projects have a firm completion date whereby the project must be completed. Under these circumstances, it is imperative to work backwards in order to determine date when project must start. Bear in mind that your career should not rely on a single project but rather on a number of projects.

Purpose

Projects are undertaken to achieve something. This implies that an IT project could generate a number of results such as a system, a software package or a recommendation based on study. Thus, the goals of projects have got to produce something tangible plus valuable to the organisation. A project needs to be goal-driven in terms of defining the work to be done, its schedule and its budget plus providing the team with a clear direction.

Since it sets out the expectations that will directly influence client's satisfaction level, projects' goals must be clearly defined plus agreed upon. The definition of PM stipulates that project activities must meet or exceed stakeholder needs and expectations. Thus, these expectations and needs cannot be met if the project's goals are not achieved. Therefore, it is vital that a project should only be undertaken to offer some sort of value to the organisation. So a specific and measurable project goal can be evaluated after a project is completed.

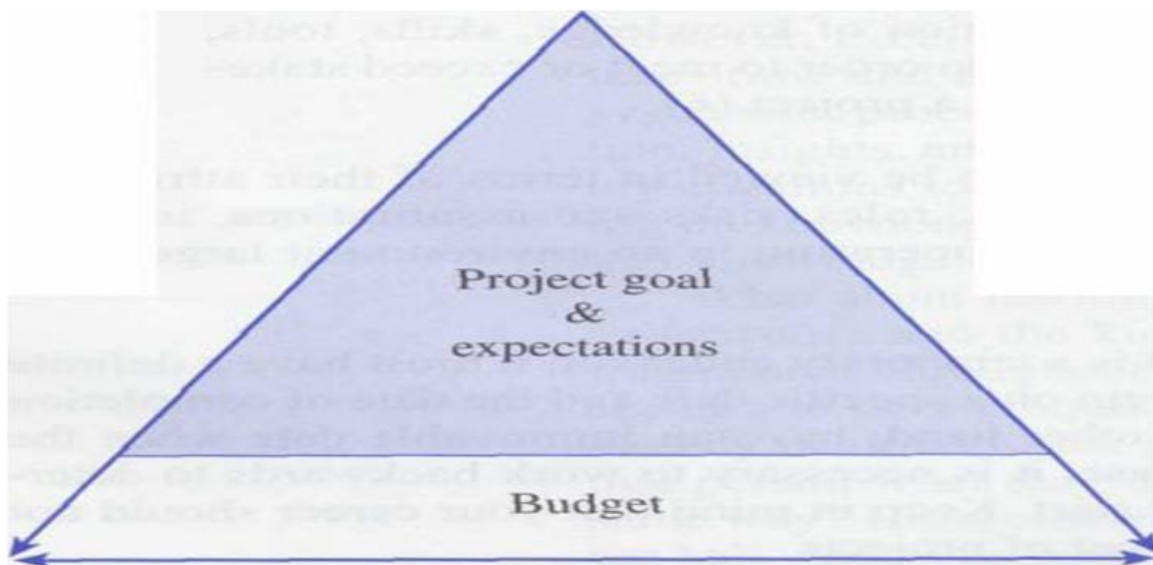
Ownership

A project must offer something of value to an individual or group who will own the project's product after it is completed. Deciding who owns this project is not effortless e.g. different groups might fight over who does or does not own the system, the data, the support and final cost of implementation and system maintenance. Even though a project may have several stakeholders i.e. people or groups who have vested interest in project outcome, any project should have a clearly defined sponsor. The sponsor could be an end user, or the client who has the capability and desire to provide direction, funding and other resources to the projects.

Resources

IT projects require time, money, people and technology. Resources offer the means for achieving a project's goal plus also functioning as a constraint e.g. the project's scope or work to be accomplished is determined solely by the project's goal i.e. if we know what we have to accomplish, then we can figure out how to accomplish it. When a project sponsor asks that an extra feature be added to system, this request will no doubt require additional resources in terms of more work on the project team's part. Thus use of project resource has an associated cost that must be included in overall project cost.

In the past, computer technology was relatively more expensive than labour needed to develop a system. But today, the labour to build a system is relatively more expensive than technology. As IT salaries increase, the cost of IT projects will become more expensive. Thus, if team members must do additional work, their time plus cost associated with the time spent doing unscheduled work must be added to project's schedule and budget. This implies that if the scope increases, the schedule and budget of the project must increase. If project's schedule and resources are fixed, the only way to decrease cost or schedule of project may be to reduce scope. The scope, schedule and budget must stay in some sort of balance to support a specific project goal. This relationship, sometimes referred to as the triple constraint is illustrated in Fig 1.1 (Marchewka, 2003).



Source: Marchewka, 2003

This is useful when making a decision that affects project's goal, scope or budget.

Roles Today

IT projects require different individuals with different skill sets. Although these skills may be different on various projects, a typical project may include the following:

Project Manager: Project manager is team leader who is responsible for ensuring that all of the project management and technical development processes are in place and are being carried out within a set of specific requirements, defined processes and quality standards.

Project Sponsor: It may be a client, customer or organisational manager who acts as a champion for the project plus provides organisational resources and direction when needed.

Subject Matter Expert(s) (SME): SME may be a user who has specific knowledge, expertise or insight in a specific functional area required to support project e.g. if an organisation wishes to develop a system to support tax decisions, having a tax expert on project team who can share his/her knowledge will be more productive than having technical people trying to learn everything about tax accounting while developing system

Technical Expert (TE): Technical expertise is required to provide technical solution to an organisational problem. TEs can include systems analysts, network specialists, programmers, graphic artists, trainers and so forth.

Irrespective of their job title, these individuals are responsible for: defining, creating and implementing the technical and organisational infrastructure to support the product of IT project.

Risks and Assumptions

All projects have an element of risk but some projects entail greater risk than others. Risk can crop up from a number of sources - this includes both internal and external to the team e.g. internal risks could crop up from estimation process or from the fact that a key member of project team might leave in middle of project. While external risks could arise from relying on other contractors or vendors. Assumptions are what we deploy to estimate the scope, schedule and budget and to access the risks associated with project. There are several unknown variables associated with projects and it is vital to identify plus make clear all of the risks and assumptions that could impact on IT projects.

Independent Tasks

Project work requires numerous independent tasks e.g. a network cannot be installed unless the hardware is delivered or certain requirements cannot be incorporated in the design until a key user is interviewed. Sometimes the delay of a single task could affect other subsequent independent tasks. Thus, a project's schedule may slip and project may not meet its planned deadline.

Organisational Change

Projects signify planned organisational change. So change must be understood and carefully managed. Since the implementation of IT project will change the way people work, the potential for resistance therefore exists and a data system which is a technical success could end up being an organisational failure.

Operating in an Environment Larger than the Project Itself

Organisations choose projects for a number of reasons and the projects chosen can impact on the organisation (Laudon & Laudon, 2006). It is vital that the project manager and team understand company's culture, environment, politics and the like. Organisational variables influence project selection, IT infrastructure, role of IT within organisation e.g. small family owned manufacturing company could have a totally different corporate culture, strategy and structure than a start-up e-commerce one.

As such projects selected, technical infrastructure and IT role for each organisation will be different. So the project team must understand both technical and organisational variables so that projects can be aligned properly with structure and strategy of organisation. Furthermore, understanding organisational variables can help the project team understand political climate within organisation plus identify potential risks and issues that could hamper project.

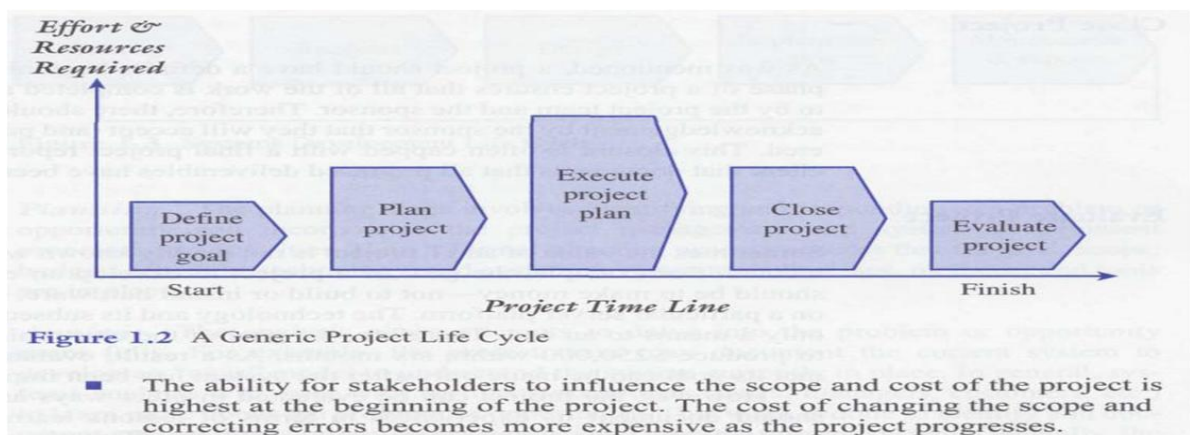
Project Life Cycle (PLC) and IT Development

Project life cycle (PLC) is a collection of logical stages or phases that maps the life of a project from its beginning to its end in order to define, build, and deliver the product of a project – that is the Information System (Marchewka, 2003). Each phase should offer one or more deliverables.

A deliverable is a tangible and verifiable product of work i.e. project plan, design specifications, delivered system, etc. Deliverables at the end of each phase equally offer real benefits through project and serve to define work and resources needed for each phase. Projects should be broken down into phases to make projects manageable plus to reduce risk.

Phase exits, stage gates or kill points are phase-end review of key deliverables that permit an organisation to evaluate a project's performance plus take immediate action to rectify any errors or problems. Even though deliverables at end of a phase can normally be approved before proceeding to the next stage, fast tracking or starting the next phase before approval is obtained can sometimes decrease project's schedule. Overlapping of phases can be risk and should only be done if risk is deemed adequate. Like all living things, projects have life cycles where they are born, grow, peak, decline and then terminate (Meredith & Mantel, 2000).

Although project life cycles may differ depending upon the industry, or project, all project life cycles will have a beginning, a middle, and an end (Gido & Clements, 1999). Fig 1.2 depicts a generic project life cycle that portrays the common phases or stages shared by most projects



Define Project Goal

Defining overall project goal should be first stage or measure of the project. The goal should focus on offering business value to organisation. Well-defined goal provides team with clear focus plus drives other project phases. In addition, several projects seem to share the following characteristics:

- The effort in terms of cost and staffing levels is low at start of project but then increases as project work is being carried out before decreasing at the end as project is completed.
- Risk and uncertainty are highest at start of project. Once the project goal is defined plus project processes, success prospect should increase

Plan Project

Once project's goal has been defined, developing a project plan is a much easier task. Project plan in effect addresses the following questions:

- What are we going to do?
- Why are we going to do it?
- How are we going to do it?
- Who is going to be involved?
- How long will it take?
- How much will it cost?
- What can go wrong and what can we do about it?
- Why did we estimate the schedule and budget?
- How will we know if we are successful?

In addition, deliverables, tasks, resources and time to complete every task must be defined for each project phase. This initial plan is called baseline plan. Define agreed upon scope, schedule and budget and it is deployed to gauge project's performance throughout the life cycle.

Execute Project Plan

After a project's goal and plan have been defined, it is time to put the plan into action. As work on project progresses, scope, schedule and budget need to be actively managed to ensure that the project accomplishes its goal. The project's progress must be documented and compared to project's baseline plan. Also, project performance must be communicated to all project stakeholders. The project team implements or delivers a complete product to organisation at the end of this phase.

Close Project

As earlier mentioned, a project should have a definite beginning and an end. The closing phase of a project certifies that all of the work is completed as planned and as agreed to by the project team and sponsor. Thus there should be some sort of acknowledgement by sponsor that they will accept (and pay for!) the project delivered.

Closure is often capped with a final project report and presentation to the client which documents that all promised deliverables promised to have been completed as specified.

Evaluate Project

Quite often, the value of IT project is not readily known when system is implemented e.g. the goal of a project to develop an e-commerce site should be to make money but not to build or install hardware, software, and web pages on a specific server platform. Technology and its consequent implementation are simply a means to an end. So the goal of the e-commerce site may be to generate 2.500.000 CFA within 6 months. As such evaluating whether project met its goal can be made only after the system has been implemented. Nonetheless, a project can be evaluated in other ways as well.

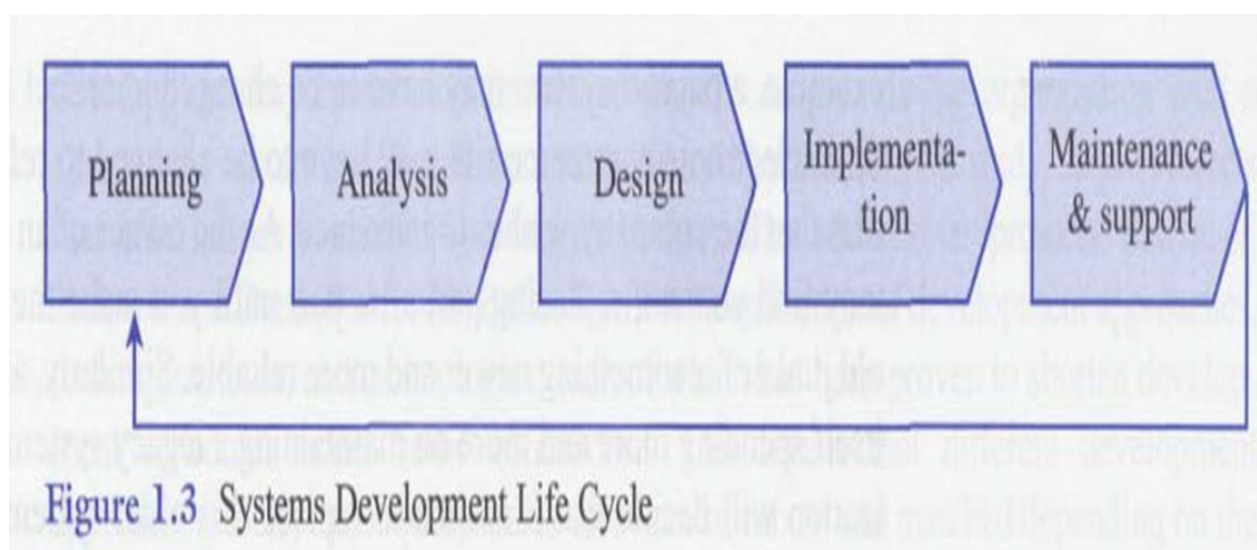
Project team should document its experience in terms of lessons learned. The things it would do the same and those that it would differently on next project based on current project experiences. These should be documented electronically plus stored across the organisation. Subsequently, many of these experiences can be translated into best practices and integrated into future projects. Additionally, both project team and project itself should be evaluated at the end of project. The project manager might evaluate performance of each team member in order to provide feedback and as part of organisation's established merit and pay raise processes and procedures.

Most often, an outside third party such as a senior manager or partner may audit project to determine if project was well-managed, provided promised deliverables, followed established processes plus met specific quality standards. Project team and project manager may also be evaluated in terms of whether they acted in a professional and ethical manner.

IT Product Life Cycle

Even though projects follow a project life cycle, IS development follows a product life cycle. Most common product life cycle in IT is Systems Development Life Cycle (SDLC) which embodies chronological phases or stages and IS follows throughout its useful life.

SDLC establishes a logical order in which system development activities occur and indicates whether to proceed from one system development activity to the next (McConnell, 1996). Although there is no universally acceptable version of SDLC, the life cycle depicted in Fig 1.3 includes generally accepted activities and phases associated with systems development.



Bear in mind that these concepts are usually covered in great detail in system analysis and design books plus courses. So for some, this may be a quick review while for others it will provide a background for understanding how IT PM and IS development activities support one another.

Planning, analysis, design, implementation and maintenance and support are 5 classic phases in systems development life cycle.

Planning

The planning stage involves identifying plus responding to a problem or opportunity as well as incorporating PM and system development processes and activities. Here proper planning process ensures that the goal, scope, budget, schedule, technology and system development plus processes, methods, and tools are in place.

Analysis

The analysis phase tries to research into problem or opportunity more fully e.g. project team may document current system to develop an “as is” model to comprehend the system currently in place. Usually, systems analysts will meet with various stakeholders (users, managers, customers, etc) to find out more/gain more knowledge about problem or opportunity. Work done to determine and document any problems or bottlenecks associated with current system, usually the “as is” analysis is followed by a requirements analysis. So specific needs and requirements for new system are identified plus documented.

Requirements can be developed through a number of ways:

- Interviewing, joint application development (JAD), conducting surveys,
- Observing work processes and reading company reports
- Deploying process-oriented, data-oriented and or object-oriented modeling techniques

The current system, user requirements and logical design of future system called the “to be” system are represented and documented (Dennis & Haley, 2000).

Design

During system design phase, project team utilises requirements and “to be” logical input models for designing the architecture to support the new IS. Architecture includes designing the network, hardware configuration, databases, user interface and application programs.

Implementation

Implementation includes the development or the construction of the system, testing and installation. Also, training, support and documentation must be in place. Although maintenance and support may not be a true phase of current projects, it is still a vital consideration.

Once a system has been implemented, it is said to be in production. Changes to the system, in the form of maintenance and enhancements are often requested to fix any discovered errors i.e. bugs that are discovered within the system to include any features that were not incorporated into the original design or even to adjust a changing business environment. Support in terms of a call centre or help desk may usually be in place in order to help users on as-needed basis. **Finally, the**

system becomes part of an organisational infrastructure plus becomes known as a legacy system. At this juncture, the system becomes very similar to a new car. Let's presuppose that you buy a new car. Over time, the car becomes less and less new as parts have to be replaced as they wear out. Even though, a system doesn't wear out like a car, changes to the system are needed as the organisation changes e.g. a payroll system may have to be changed to reflect changes in tax code or an e-commerce site may have to be changed to reflect a new line of products the company wishes to introduce.

So as an owner of an old car or a classic one, you may find yourself replacing part after part until you make a decision to trade in the old Junker for something newer and more reliable. Likewise, an organisation may find itself spending more and more on maintaining a legacy system. Eventually organisation will decide that it's time to replace this older system with a newer one that will be more reliable, require less maintenance plus better meets its needs. Thus, under these circumstances, a new life cycle begins putting SDLC into Practice.